A machine learning–based approach to automatically detect lymphoproliferative disorders using flow data collected from the BD OneFlow™ LST panel

Yu-Fen Wang[1,2], Imelda Omana-Zapata[3], Josh Zollet[3], Teng-Jen Chang[1,2], Yun-Chun Huang[1,2], Bor-Sheng Ko[1,2,4,5]

Affiliations: [1]AHEAD Medicine Corporation, Berkeley, CA, U.S.; [2]AHEAD Intelligence Ltd, Taipei, Taiwan; [3]BD Biosciences, San Jose, California, U.S.; [4]Department of Hematological Oncology. National Taiwan University Cancer Center, Taipei, Taiwan; [5]Department of Internal Medicine, National Taiwan University Hospital, Taipei, Taiwan

Introduction

Automated analysis solution is an emerging demand due to the growing complexity and capacity of flow cytometry.

This study develops a machine learning based analysis tool for automated aid to the BD OneFlow™ LST reagent, distinguishing normal or abnormal cell phenotypes peripheral blood (PB) specimens.

Methods

A subset of 176 fcs files from the PB samples from BD OneFlow™ LST study was used (Table 1). The BD OneFlow™ Lymphoid Screening Tube (LST) reagent is an 8-color assay that identifies the presence of lymphoid neoplastic cells and determines the lymphocyte lineage. The OneFlow™ LST Panel is listed in Table 2.

Table 1. Demographics by Site, Instrument Type, and Label for Model Training

Study Site	Instrument		Label	
	FACSCanto™II	FACSLyric™	Normal	Abnormal
1	21	21	24	18
2	25	25	34	16
3	22	22	34	10
4	20	20	16	24
Total	**88**	**88**	**108**	**68**

Abnormal: PB samples with B-cell abnormalities

Table 2. BD OneFlow™ LST Panel Composition

Laser								
Fluorochrome	BD Horizon™ V450	BD Horizon™ V500	FITC	PE	PerCP-Cy5.5	PE-Cy7	APC	APC-H7
Antibody	CD20 & CD4	CD45	CD8 & Igλ	CD56 & Igκ	CD5	CD19 & TCRγδ-1	CD3	CD38

The previously published classification model training experiment and model training framework design (Figure 1a) was used and a 5-fold cross validation (Figure 1b) was used to evaluate performance.

The speed of automated analysis by the deployed model was evaluated by measuring the average time from upload fcs file to prediction result displayed on the user interface.

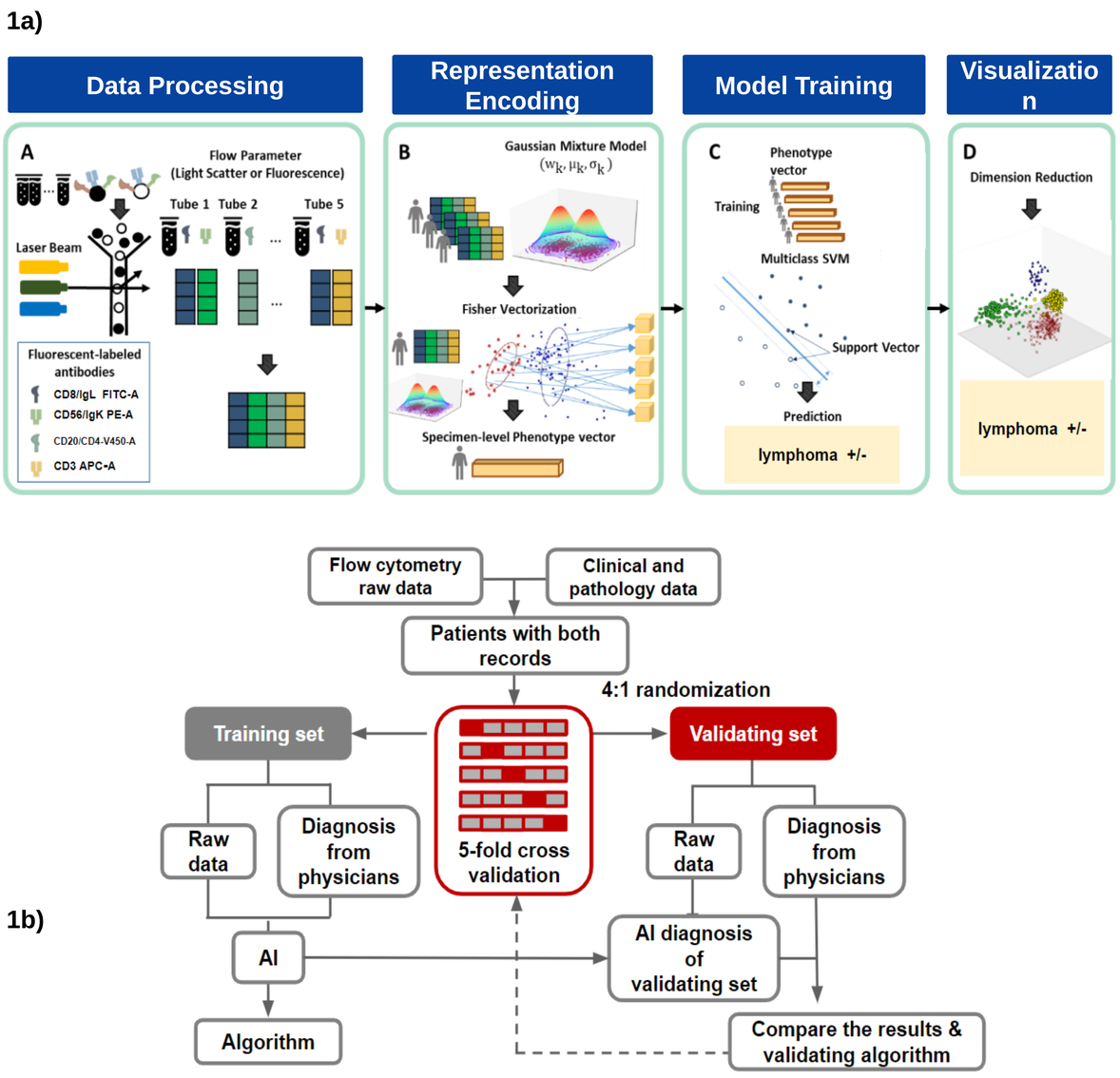

Figure 1. Model Development Scheme

1a) Machine Learning Framework Design

1b) Workflow of development classification model using 5-fold cross validation

The overall design was previously utilized in acute leukemia subtype classification model training using data from UPMC and was published at AJCP (2021) doi: 10.1093/AJCP/AQAB148.

Results

The abnormality classification model achieved an average AUC of 97.1% and an average accuracy of 92.6%. Specificity and sensitivity were 93.5% and 91.0%, respectively (Table 2). By adjusting the abnormality detection threshold along the ROC curve, the false negative rate reduced from 19% to 7% while the accuracy slight reduced from 91.5% to 90.3% (Figure 2). The deployed model used a threshold with specificity of 93.5% and sensitivity of 91.0% to the AHEAD cloud platform.

Contact

AHEAD: Andrea.Wang@aheadmedicine.com

BD: Imelda_omana-zapata@bd.com

Trial access of AHEAD platform

AHEAD's Publications

Results

Table 3. Model Performance Summary

Performance	5 Fold Average (%)
AUC	97.1
Accuracy	92.6
Specificity	93.5
Sensitivity	91.0
False Positive Rate	5.5
False Negative Rate	10.2

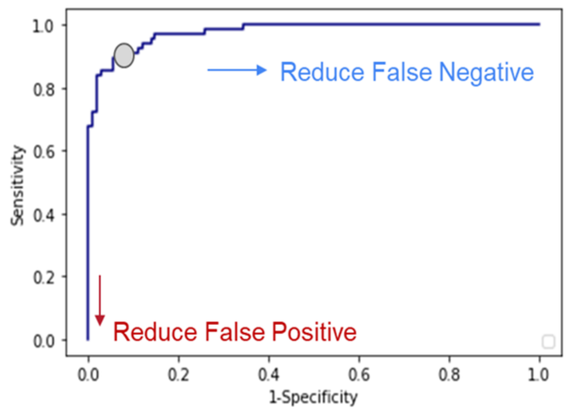

Figure 2. AUC graph of the LST model

The model threshold could be adjusted along the ROC curve to optimize toward lower false positive or false negative rate before been deployed to AHEAD cloud platform

The prediction results displayed includes an interactive 3-dimensional plot to compare the sample to the training database at specimen level and 4 interactive scatter plots showing predicted T, B and NK cells of the sample in question.

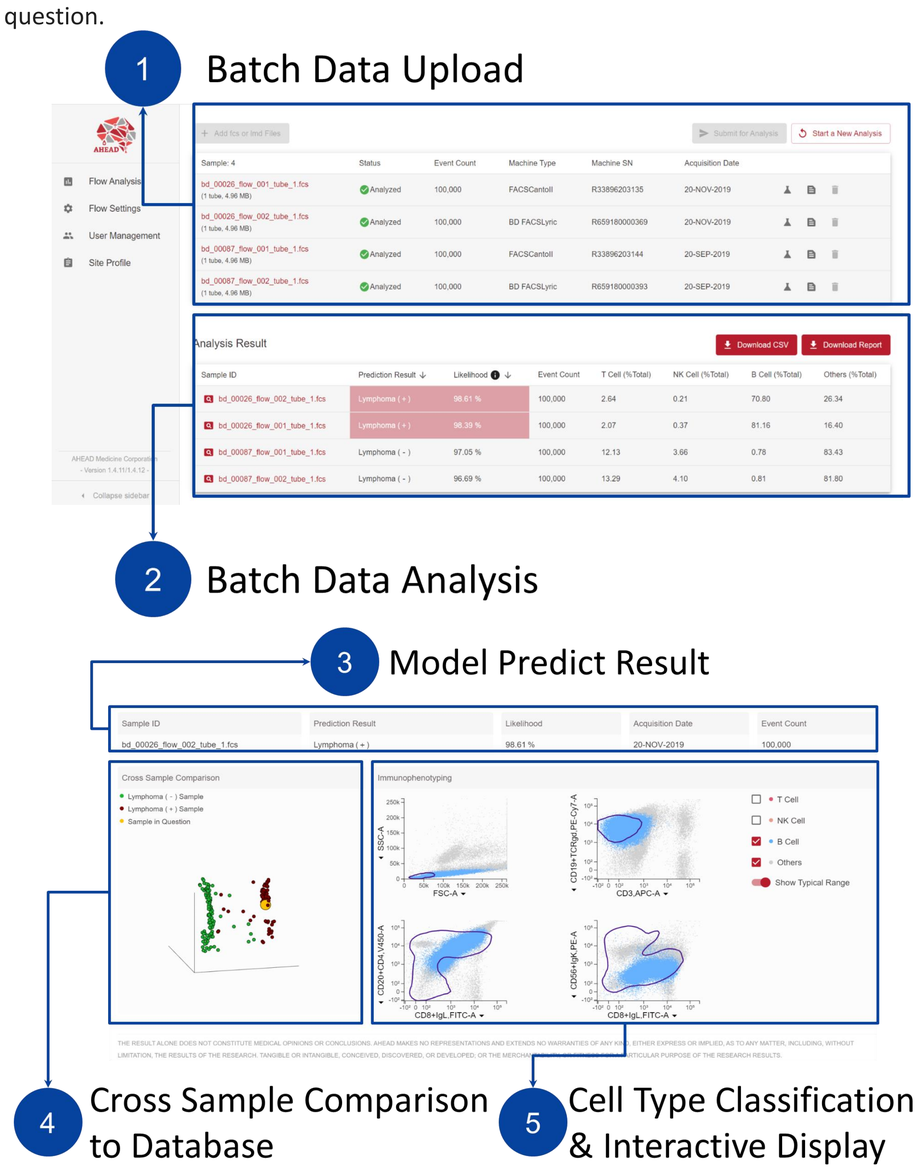

Figure 3. AHEAD Cloud-based LST model Key Features

The LST model is currently been deployed to the cloud and support automated analysis on flow data acquired on OneFlow™ LST. Currently, the model only support detection of B-cell abnormality on PB specimen with 91% sensitivity and 93.5% specificity based on the model performance evaluation using 5 fold cross validation.

Table 3. Average Time from fcs upload to prediction result displayed on AHEAD cloud platform

Number of OneFlow LST test fcs file	Time
1 fcs	21.3 seconds
5 fcs	47.3 seconds

Conclusion

- This study has achieved great performance in identifying B-cell abnormality in PB specimens using data across 4 sites and 2 instrument types.
- The cloud-based model can perform automated specimen level abnormality detection and lymphocyte classification less than a minute per sample.

Next Step

- Expanding training dataset to support the prediction of other types of abnormalities in the lymphoma screening test
- Establishing new use cases in hematology studies

For Research Use Only. Not for use in diagnostic or therapeutic procedures.

BD OneFlow™ LST is not available for sale in the U.S. and the use of AI to analyze data is not an approved use in countries where CE market OneFlow™ LST is registered.
BD flow cytometers are Class 1 Laser Products
In the U.S., the BD FACSLyric™ Flow Cytometer is for In Vitro Diagnostic Use with BD FACSuite™ Clinical Application for up to six colors.
In the U.S., the BD FACSLyric™ Flow Cytometer is for Research Use Only with BD FACSuite™ Application for up to 12 colors. Not for use in diagnostic or therapeutic procedures.
In the E.U.., the BD FACSLyric™ Flow Cytometer with the BD FACSuite™ Clinical and BD FACSuite™ Applications is an in vitro diagnostic medical device bearing a CE mark.
In the U.S., the BD FACSCanto™ II Flow Cytometer is for In Vitro Diagnostic Use for up to six colors. Seven and eight colors are for Research Use Only.
In the E.U., the BD FACSCanto™ II Flow Cytometer is no longer available for sale.
In the E.U., the BD OneFlow LST is CE marked in compliance with the EU IVD Regulation (2017/746).

BD, the BD Logo, BD FACSLyric, BD OneFlow, FACSCanto and Horizon are trademarks of Becton, Dickinson and Company or its affiliates. All other trademarks are the property of their respective owners.
BD-100797 (v1.0) 0923